

# Take-Home Assignment

*From conversation to shipped internal tool: designing an autonomous AI implementation agent*

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| <b>Role</b>             | AI Implementation Intern                                                      |
| <b>Estimated effort</b> | 6–8 hours (spread over up to 5 days)                                          |
| <b>What you submit</b>  | A written design document and an architecture / flow diagram                  |
| <b>How we assess it</b> | Depth of thinking, clarity, and sound product & systems judgment — not polish |

## 1. Overview

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Our AI implementation team exists to solve internal problems with AI: automating the manual, repetitive work that slows other teams down, and turning it into workflows, agents, and small apps that people across the company actually use. Good candidate use cases surface constantly — in team calls, operational reviews, and support or process discussions — but a lot of value leaks out between someone describing a painful workflow and a working tool that fixes it. The bottleneck is rarely a shortage of ideas; it is the manual effort of turning a raw signal into a defined problem, then into a spec, then into something a colleague can actually use.

Your challenge is to design an AI system that closes that gap as autonomously as responsibly possible. Concretely, the system should be able to identify a worthwhile internal use case on its own, turn it into a product requirements document (PRD), produce a mock-up or wireframe of the proposed solution, and then drive the build of a working internal app — with a human stepping in at the right moments to steer and approve.

You will build this on an internal low-code AI platform that lets the team assemble workflows, autonomous agents, and deployable apps. You do not need to know the specifics of that platform. Assume it can run multi-step agentic workflows, call tools and external services, hold state, request human input mid-run, and deploy a simple app at the end. Where a capability matters to your design, state the assumption you are making about it and move on.

## 2. The Challenge

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Design — on paper — an autonomous, agentic system that takes an internal use case all the way from discovery to a shipped app. The end-to-end flow you are designing for is:

1. **Discover** — the system surfaces a candidate internal use case worth solving, from whatever raw inputs you decide it should draw on.
2. **Define** — it drafts a PRD for that use case: the problem, the internal users, scope, requirements, and success criteria.

3. **Design** — it produces a mock-up or wireframe of the proposed solution.
4. **Build** — it uses the approved PRD and design to build a working internal app on the platform.

*The system should operate autonomously by default, pausing for a human only where human judgment genuinely adds value or reduces risk. Deciding where those moments are is a core part of the assignment, not an afterthought.*

## 3. Questions Your Design Must Answer

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We care far more about how you reason through the following than about any single “correct” answer. Address each explicitly.

### 3.1 Discovery

- Where do candidate use cases come from? What raw inputs or signals does the system draw on, and why those?
- How does the system decide a use case is worth pursuing? What makes something a real problem versus noise — and how do you prioritise when several surface at once?
- How does discovery actually run — on a trigger, on a schedule, continuously? Who or what kicks it off?

### 3.2 The Workflow

- Once a use case is chosen, what are the end-to-end steps from that point to a shipped app? Lay out the full pipeline.
- What are the discrete agents or steps, what does each one take in and hand off, and how does state carry across them?
- What happens when a step produces something wrong or low quality — how does the system catch it, and what does it do next?
- What does the system do when it is genuinely uncertain or blocked, rather than pushing ahead on a bad assumption?

### 3.3 Human-in-the-Loop (HITL)

- Where exactly does a human need to be in the loop, and why is human judgment needed at each of those points specifically?
- What is the human being asked to do at each checkpoint — approve, edit, choose between options, or provide missing context?
- How does the system behave while it waits, and how does it incorporate the feedback it gets back?
- How do you keep the human from becoming a bottleneck while still preventing the agent from shipping the wrong thing?

### 3.4 Trade-offs, Risks & Assumptions

- What are the main failure modes, and how does your design reduce or contain them?

- What assumptions are you making about the platform, the inputs, or the users? State them plainly.
- What did you deliberately leave out of scope, and why?
- How would you know the system is working — what would you measure?

## 4. What to Submit

| Deliverable                        | What we expect                                                                                                                                                                                                          |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Design document</b>             | A written document (roughly 3–6 pages, any format) that answers the questions in Section 3 and walks through your end-to-end design. Prose, not just bullet points — we want your reasoning, not only your conclusions. |
| <b>Architecture / flow diagram</b> | A visual of the system: the steps or agents, how data and control flow between them, and — clearly marked — where the human-in-the-loop checkpoints sit. Any tool is fine; legibility is what matters.                  |

*You are encouraged to make and state assumptions wherever the brief is open. There is no hidden “right” architecture — a clearly reasoned design built on explicit assumptions is exactly what we are looking for.*

**Bonus (optional).** Record a short video (5–8 minutes) walking us through your end-to-end autonomous system — how discovery kicks off, how the workflow moves from use case to shipped app, and where the human-in-the-loop checkpoints sit and why. A screen recording over your diagram is perfect; we want to hear you reason through it, not a polished production. Share it as a link (Loom, Drive, or similar).

## 5. How We Evaluate

| Criterion                   | What good looks like                                                                       |
|-----------------------------|--------------------------------------------------------------------------------------------|
| <b>Problem framing</b>      | You identify the real problem and scope it sensibly before jumping to a solution.          |
| <b>Systems thinking</b>     | The workflow is coherent end to end; steps, hand-offs, and state make sense.               |
| <b>HITL judgment</b>        | Human checkpoints are placed where they genuinely matter — neither too many nor too few.   |
| <b>Handling uncertainty</b> | The design degrades gracefully: it catches errors and knows when to ask rather than guess. |
| <b>Communication</b>        | The document and diagram are clear, well-organised, and easy to follow.                    |

## 6. Logistics & Submission

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- Spend no more than a day of focused effort; we are not looking for a production system.
- Use any tools you like, including AI assistants — be ready to explain and defend your choices.
- Submit your design document and diagram as a single PDF, or a link, to your point of contact.
- Optional: note anything you would explore next if you had more time.

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*Questions about the brief are welcome — knowing what to ask is part of the work. Good luck; we are excited to see how you think.*